Numerical reconstruction of Schrödinger equations with quadratic nonlinearities

Khaoula El Maddah

Research Unit of Applied and Computational Mathematics
Faculty of ITEE

Joint work with Teemu Tyni and Matti Lassas

University of Oulu









Nonlinear Schrödinger Equation

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with smooth boundary $\partial\Omega$ and $p \in \mathbb{N}_{\geq 2}$. We consider the semilinear elliptic boundary value problem

$$\begin{cases} -\Delta u + q\, u^p = 0 & \text{in } \Omega, \\ u = f_b(x) & \text{on } \partial\Omega, \end{cases}$$

The semilinear elliptic equation admits a unique solution $u \in C^{2,\alpha}(\overline{\Omega})$ with $0 < \alpha < 1$, when $f_b \in C^{2,\alpha}(\partial\overline{\Omega})$ is sufficiently small.

UNIVERSITY OF OULU

Notation

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with smooth boundary $\partial\Omega$ and $p \in \mathbb{N}_{\geq 2}$. We consider the semilinear elliptic boundary value problem

$$\begin{cases} -\Delta u + q\, u^p = 0 & \text{in } \Omega, \\ u = f_b(x) & \text{on } \partial\Omega \end{cases}$$

The Dirichlet to Neumann map (DN map) is defined by

$$\begin{aligned} \Lambda_q : \; C^{2,\alpha}(\partial\Omega) &\longrightarrow C^{1,\alpha}(\partial\Omega) \\ f &\longmapsto \partial_\nu u\big|_{\partial\Omega} \end{aligned}$$

where ν is the outward normal vector to Ω.

UNIVERSITY OF OULU

Well-posedness via Banach's Fixed Point Theorem

By applying **Banach Fixed Point Theorem**, for any initial point $u_0 \in \bar{B}_\delta(0)$, the sequence $\{u_n\}$ defined by:

$$u_n = T(u_{n-1}), \quad n \geq 1,$$

converges to a fixed point u^*, where

$$T : C^{2,\alpha} \to C^{2,\alpha} \qquad \Phi : C^{\alpha} \to C^{2,\alpha}$$
$$v \longrightarrow \Phi(qv^p) \qquad F \longrightarrow u_F$$

where u_F and u^* satisfy:

$$\begin{cases} -\Delta u_F + F = 0 & \text{in } \Omega, \\ u_F = \epsilon f & \text{on } \partial\Omega \end{cases} \qquad \begin{cases} -\Delta u^* + q(u^*)^p = 0 & \text{in } \Omega, \\ u^* = \epsilon f & \text{on } \partial\Omega \end{cases}$$

The iterative process satisfies the following system:

$$\begin{cases} -\Delta u_n + q u_{n-1}^2 = 0 & \text{in } \Omega, \\ u_n = \epsilon f & \text{on } \partial\Omega. \end{cases}$$

UNIVERSITY OF OULU

Theorem (Newton well-posedness and convergence)

Let $u \in V = \{u \in H^1(\Omega, \mathbb{C}) : u|_{\partial\Omega} = f_b\}$ *solve*

$$F(u)(v) = \int_\Omega \nabla u \cdot \overline{\nabla v} + q(x)\, u^p\, \overline{v}\, dx = 0, \qquad \forall\, v \in V_0.$$

(i) *For* u_n *sufficiently close to* u*, the Fréchet derivative*

$$DF(u_n)[\delta u, v] = \int_\Omega \nabla \delta u \cdot \overline{\nabla v} + p\, q(x)\, u_n^{p-1} \delta u\, \overline{v}\, dx$$

is bounded and coercive on $V_0 \times V_0$*. By Lax-Milgram, the Newton correction*

$$DF(u_n)[\delta u, v] = -F(u_n)(v)$$

admits a unique $\delta u \in V_0$*.*

(ii) *If* u_0 *is sufficiently close to* u*, the Newton iteration*

$$u_{n+1} = u_n + \delta u_n$$

is well defined and converges quadratically to the solution u*.*

The inverse problem

$$\begin{cases} -\Delta u + q\, u^p = 0 & \text{in } \Omega, \\ u = f_b(x) & \text{on } \partial\Omega \end{cases}$$

Inverse Problem

The goal is to reconstruct q from the DN map Λ_q

UNIVERSITY OF OULU

Higher order linearization method

Let $\varepsilon > 0$ be a small parameter:

$$\begin{cases} -\Delta u + q\, u^2 = 0, & x \in \Omega, \\ u = \varepsilon f, & x \in \partial\Omega. \end{cases}$$

Differentiating with respect to ε and evaluating at $\varepsilon = 0$ yields:

$$\begin{cases} -\Delta v = 0, & x \in \Omega, \\ v = f, & x \in \partial\Omega. \end{cases} \qquad \begin{cases} -\Delta w + 2\, q\, v^2 = 0, & x \in \Omega, \\ w = 0, & x \in \partial\Omega. \end{cases} \tag{1}$$

Where

$$v = \left.\frac{\partial u}{\partial \varepsilon}\right|_{\varepsilon=0}, \qquad w = \left.\frac{\partial^2 u}{\partial \varepsilon^2}\right|_{\varepsilon=0}.$$

UNIVERSITY OF OULU

Higher order linearization method

Let $\varepsilon > 0$ be a small parameter:

$$\begin{cases} -\Delta u + q\, u^2 = 0, & x \in \Omega, \\ u = \varepsilon\, f, & x \in \partial\Omega. \end{cases}$$

Differentiating with respect to ε and evaluating at $\varepsilon = 0$ yields:

$$\begin{cases} -\Delta v = 0, & x \in \Omega, \\ v = f, & x \in \partial\Omega. \end{cases} \qquad \begin{cases} -\Delta w + 2\, q\, v^2 = 0, & x \in \Omega, \\ w = 0, & x \in \partial\Omega. \end{cases} \tag{2}$$

Integrating (2) over the domain yields

$$\int_{\partial\Omega} \partial_\nu w \, \mathrm{d}s = \int_\Omega \Delta w \, \mathrm{d}x = 2 \int_\Omega q\, v^2 \, \mathrm{d}x. \tag{3}$$

Choice of the boundary data f

Let's consider two boundary data $f_+ = v_1 + v_2$ and $f_- = v_1 - v_2$. So we get two solutions u_+ and u_-.
The Polarization identity holds:

$$f_+^2 - f_-^2 = (v_1 + v_2)^2 - (v_1 - v_2)^2 = 4v_1 v_2.$$

We choose the Calderon exponentials

$$v_1(x) = \frac{1}{2} e^{(-\zeta - i\xi)\cdot x/2} \quad \text{and} \quad v_2(x) = \frac{1}{2} e^{(\zeta - i\xi)\cdot x/2}$$

Then

$$v_1 v_2 = \frac{1}{4} e^{-i\xi\cdot x}.$$

UNIVERSITY OF OULU

Formulation of Fourier transform of q

Therefore, we can obtain the *two* measurements (for each ξ):

$$\frac{d^2}{d\varepsilon^2}\int_{\partial\Omega}\partial_\nu u_+\,\mathrm{d}s = 2\int_\Omega q(v_1+v_2)^2\,\mathrm{d}x$$

and

$$\frac{d^2}{d\varepsilon^2}\int_{\partial\Omega}\partial_\nu u_-\,\mathrm{d}s = 2\int_\Omega q(v_1-v_2)^2\,\mathrm{d}x,$$

then their difference is

$$\frac{d^2}{d\varepsilon^2}\int_{\partial\Omega}\partial_\nu(u_+-u_-)\,\mathrm{d}s\big|_{\varepsilon=0} = 2\int_\Omega 4qv_1v_2 = 2\int_\Omega q(x)e^{-\mathrm{i}\xi\cdot x}\,\mathrm{d}x, \qquad (4)$$

UNIVERSITY OF OULU

Formulation of Fourier transform of q

Therefore, we can obtain the *two* measurements (for each ξ):

$$\frac{d^2}{d\varepsilon^2}\int_{\partial\Omega}\partial_\nu u_+\,\mathrm{d}s = 2\int_\Omega q(v_1+v_2)^2\,\mathrm{d}x$$

and

$$\frac{d^2}{d\varepsilon^2}\int_{\partial\Omega}\partial_\nu u_-\,\mathrm{d}s = 2\int_\Omega q(v_1-v_2)^2\,\mathrm{d}x,$$

then their difference is

$$\frac{d^2}{d\varepsilon^2}\int_{\partial\Omega}(\Lambda_q(\varepsilon f_+)-\Lambda_q(\varepsilon f_-))\,\mathrm{d}s\big|_{\varepsilon=0} = 2\int_\Omega q(x)e^{-\mathrm{i}\xi\cdot x}\,\mathrm{d}x, \qquad (5)$$

UNIVERSITY OF OULU

Reconstruction of the Fourier matrix

We use a uniform Cartesian grid of square cells $p_{m,n}$ of side $h > 0$ over Ω, with

$$(m, n) \in \{0, \ldots, N_x - 1\} \times \{0, \ldots, N_y - 1\}.$$

A function $f : \Omega \to \mathbb{C}$ is approximated as

$$f(X) = \sum_{m,n} f_{m,n}\, \varphi_{m,n}(X),$$

where the indicator basis functions are

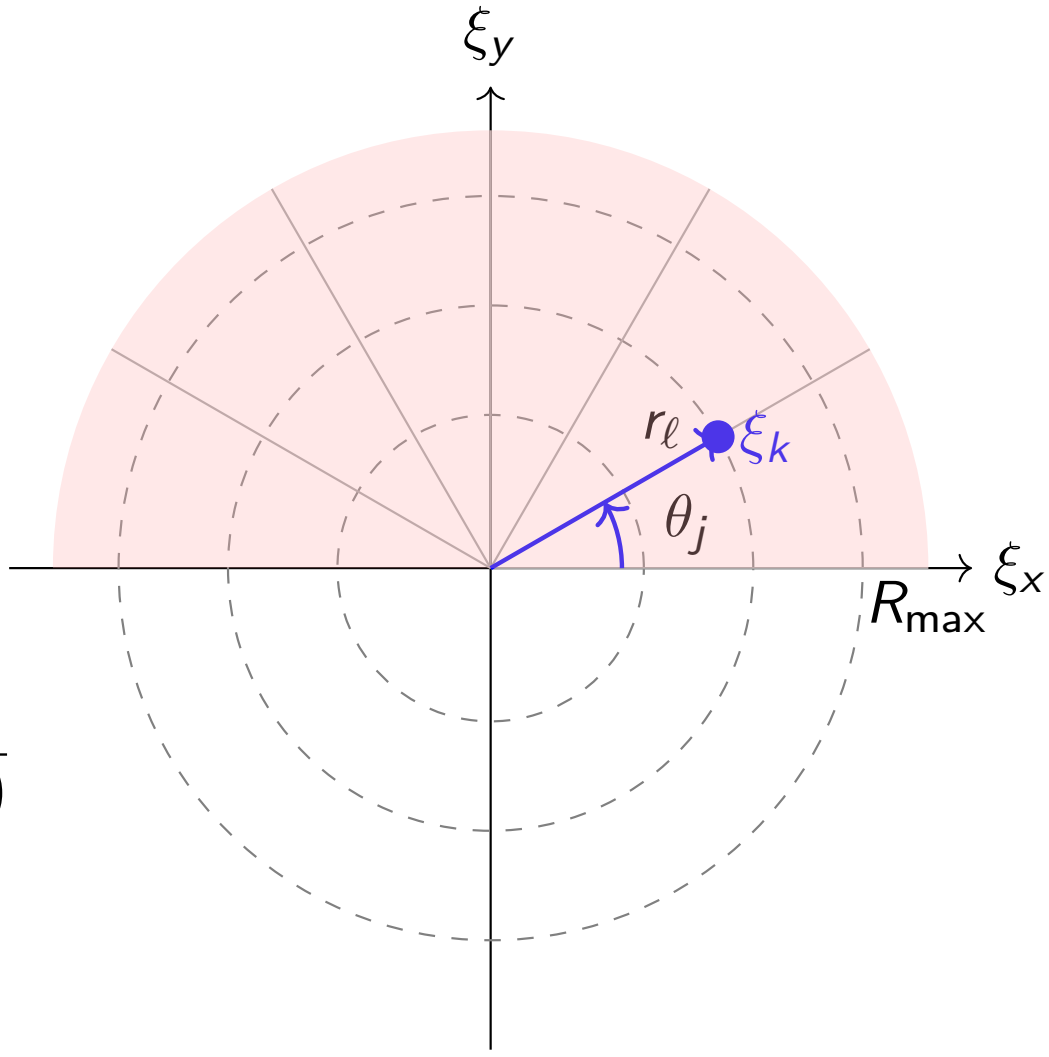
$$\varphi_{m,n}(X) = \begin{cases} 1, & X \in p_{m,n}, \\ 0, & \text{otherwise.} \end{cases}$$

For each frequency $\xi_k \in \mathbb{R}^2$,

$$\widehat{f}(\xi_k) = \sum_{m,n} f_{m,n} \int_{p_{m,n}} e^{-i\,\xi_k \cdot X}\, dX.$$

UNIVERSITY OF OULU

Sampling and Fourier Matrix



$$\widehat{q}(-\xi) = \overline{\widehat{q}(\xi)}$$

$$\xi_k = (r_\ell \cos\theta_j,\ r_\ell \sin\theta_j) = (\xi_{k,x}, \xi_{k,y})$$

Sampling and Fourier Matrix

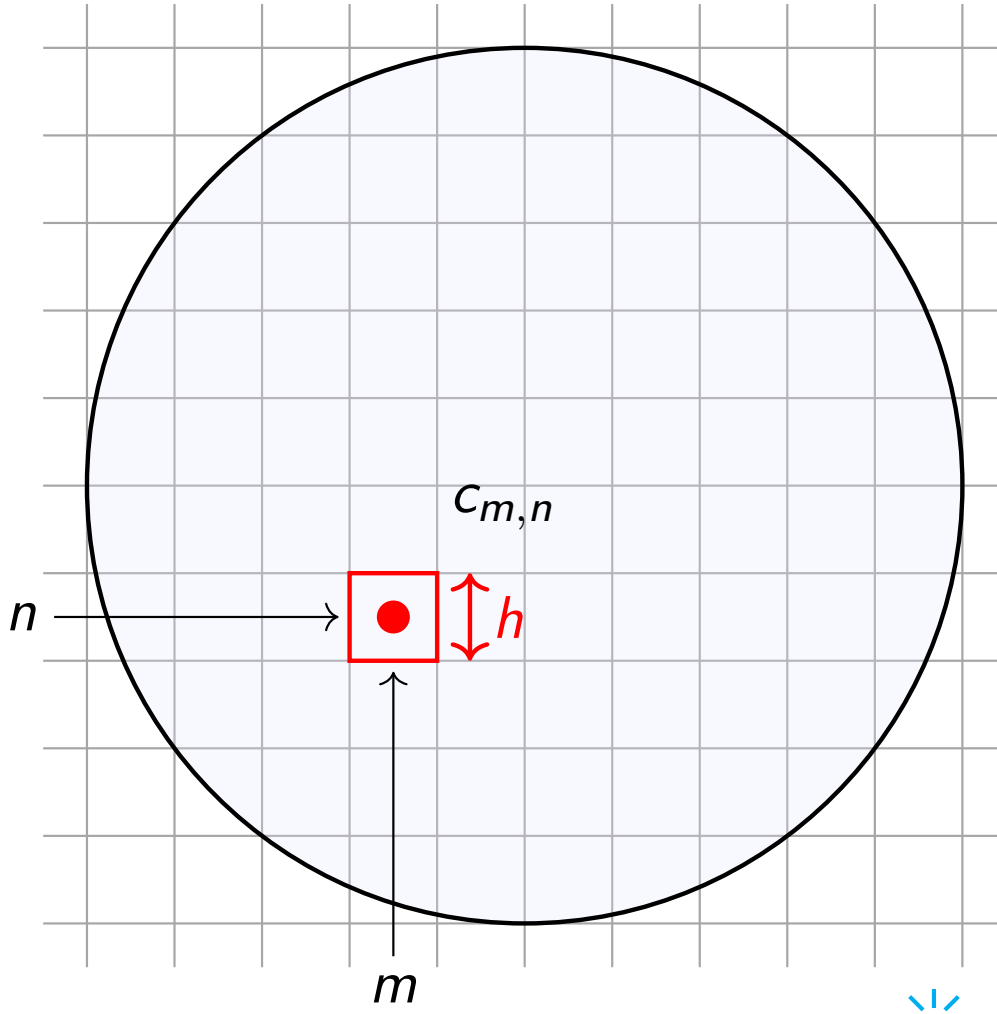
$$c_{m,x} = x_{\min} + \left(m + \tfrac{1}{2}\right) h$$

$$c_{n,y} = y_{\min} + \left(n + \tfrac{1}{2}\right) h$$

$$E_{(k_x,k_y),(m,n)} = h^2 \, \mathrm{sinc}\left(\tfrac{1}{2}\, \xi_{k,x} h\right) \times \mathrm{sinc}\left(\tfrac{1}{2}\, \xi_{k,y} h\right) e^{-i\,(\xi_{k,x} c_{m,x} + \xi_{k,y} c_{n,y})}$$

Matrix formulation:

$$F = E\, q$$



Noisy measurement

The measured Fourier data take the form

$$I_{\text{noisy}}(\varepsilon) = \int_{\partial\Omega} \partial_\nu (u_+ - u_-)_\varepsilon \, \mathrm{d}s + \eta(\xi),$$

where $\eta(\xi)$ is complex Gaussian noise.

Each frequency $\xi \in \mathbb{R}^2$ is corrupted independently:

$$\eta(\xi) \sim \mathcal{N}_{\mathbb{C}}(0, \sigma^2).$$

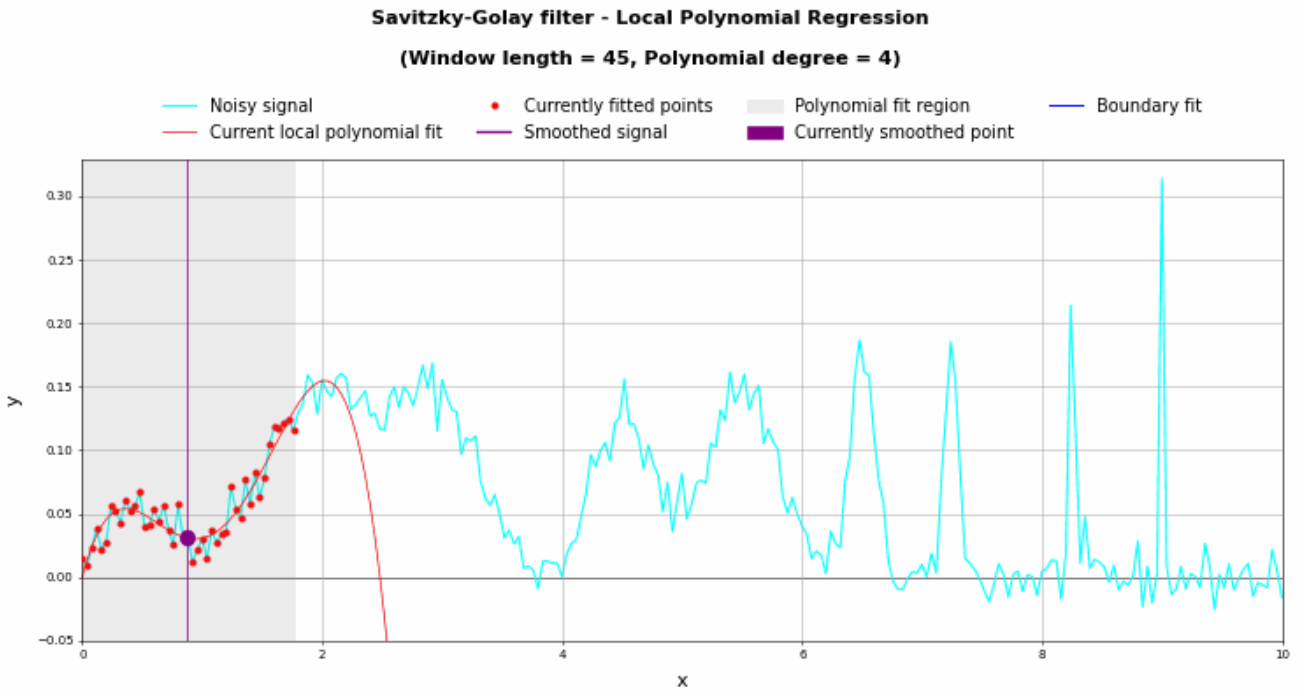
UNIVERSITY OF OULU

Savitzky–Golay Filtering

We apply SG to the noisy Fourier data to estimate higher-order derivatives stably:

$$\partial_\varepsilon^n I_{noisy}(\varepsilon)|_{\varepsilon=0} \approx \sum_{j=-J}^{J} c_j^{(n)} I_{noisy}(\varepsilon_j), \qquad \varepsilon_j = jh_\varepsilon,$$

where $c_j^{(n)}$ are the SG differentiation coefficients.



Source: Wikipedia

Regularization

Tikhonov regularization

$$q_\lambda = \arg\min_{q \in \mathbb{C}^N} \left\{ \|Eq - F\|_2^2 + \lambda \|\Gamma q\|_2^2 \right\},$$

where $\lambda > 0$ is the regularization parameter, and $\Gamma \in \mathbb{C}^{K \times N}$ is a regularization operator.
The minimizer q_λ satisfies the normal equations:

$$\left(E^\top E + \lambda \Gamma^\top \Gamma \right) q_\lambda = E^\top F.$$

Total Variation (TV) regularization

$$q_\lambda = \arg\min_{q \in \mathbb{C}^N} \left\{ \|Eq - F\|_2^2 + \lambda \, \mathrm{TV}(q) \right\},$$

where $\lambda > 0$ controls the trade-off between data fidelity and regularity, and the TV semi-norm is given by

$$\mathrm{TV}(q) = \sum_i \left(|\partial_x q_i| + |\partial_y q_i| \right).$$

UNIVERSITY OF OULU

Numerical examples

The forward problems using the `FEniCSx` framework with finite-element discretization and Newton's method on the circular domain

$$\Omega = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

- $N_\varepsilon = 64$ sampling points, uniformly distributed in $\varepsilon \in [-2, 2]$ for Savitzky–Golay differentiation of the noisy data $F(\xi_k)$
- Noise level: 1%
- regularization parameter $\lambda = 10^{-10}$ for Tikhonov and $\lambda = 10^{-8}$ for TV and the spatial grid is set to $N_x = N_y = 150$.
- The measurements are evaluated at $N_r = 60$ radial frequencies and $N_\theta = 30$ angular directions, with maximum radius $R_{\max} = 5.0$.

UNIVERSITY OF OULU

Reconstruction problem

We recover the potential q by solving the following regularized inverse problem:

$$q_{\mathrm{rec}} = \arg\min_{q} \; \frac{1}{2}\Big\| \underbrace{(\mathcal{D}^2_{\mathrm{SG}} I_{\mathrm{noisy}})(0)}_{\text{SG 2nd derivative of } I \text{ at } \varepsilon=0} - Eq \Big\|_2^2 + \lambda \mathcal{R}(q), \qquad (6)$$

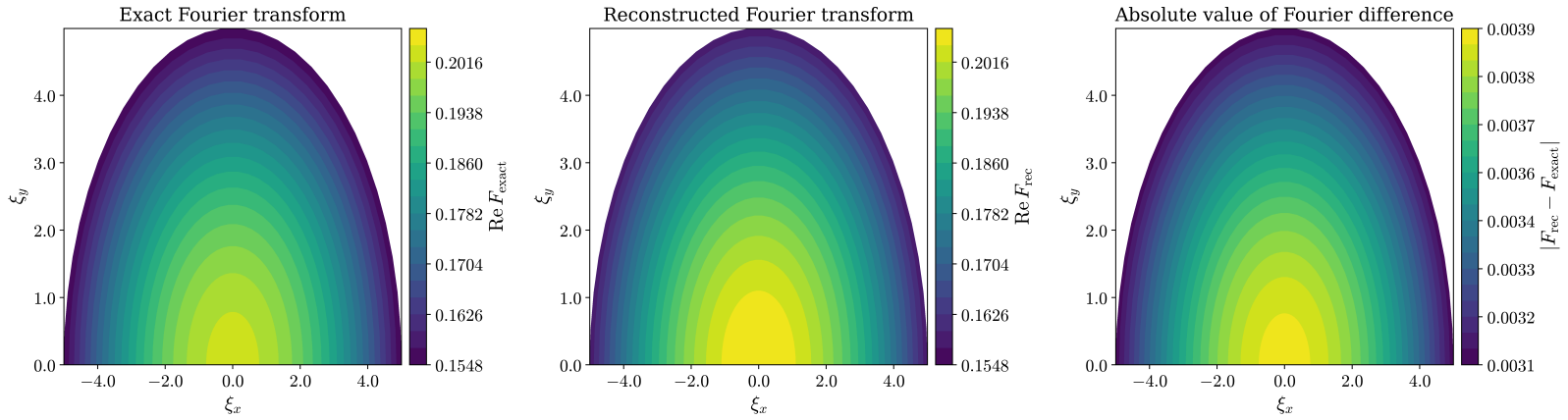
with

$$\mathcal{R}(q) \in \Big\{ \|q\|^2_{L^2(\Omega)} \quad \text{(Tikhonov)} \quad \text{or} \quad \mathrm{TV}(q) \quad \text{(Total Variation)} \Big\}.$$

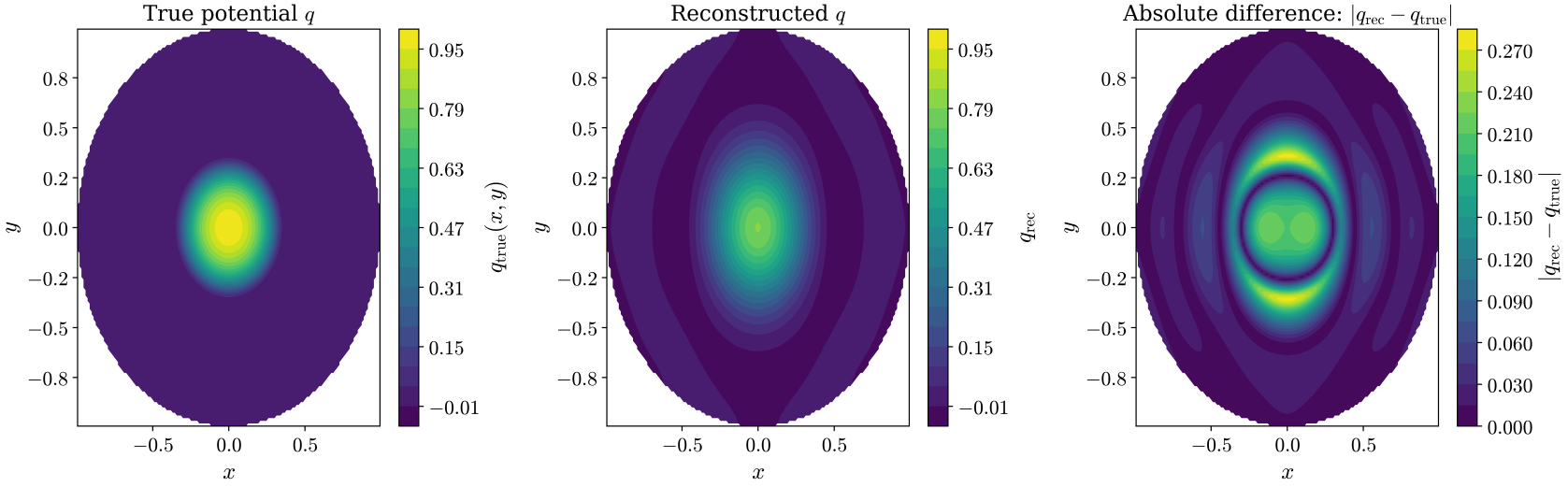
The choice of $\mathcal{R}$ depends on the desired reconstruction: smooth (Tikhonov) or edge-preserving (TV).

UNIVERSITY OF OULU

Example 1: Centered bump

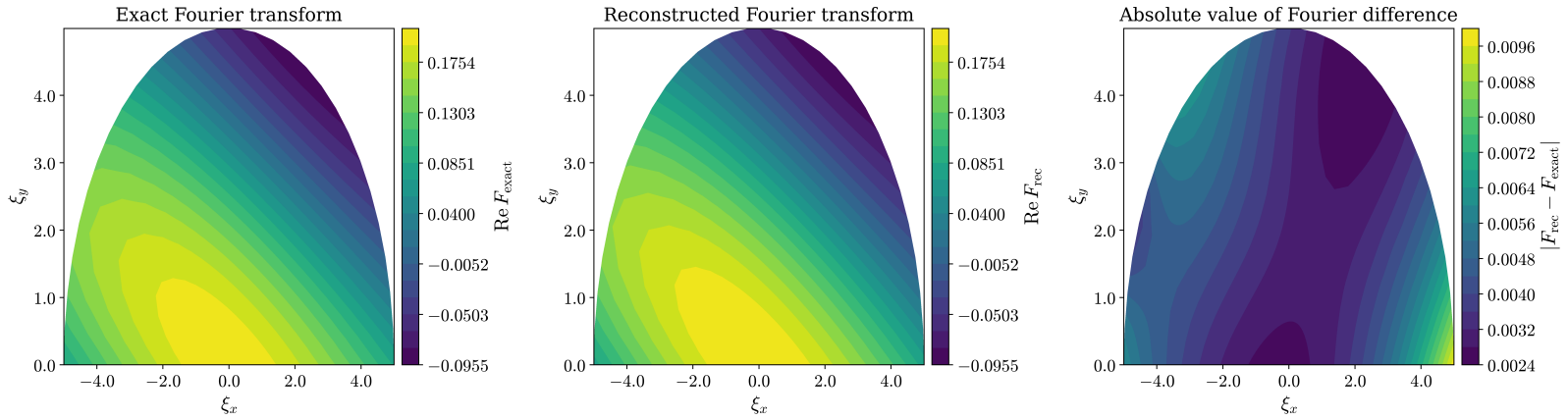


Recovered real Fourier data $\Re\,\mathcal{F}(q)$ for the centered bump.

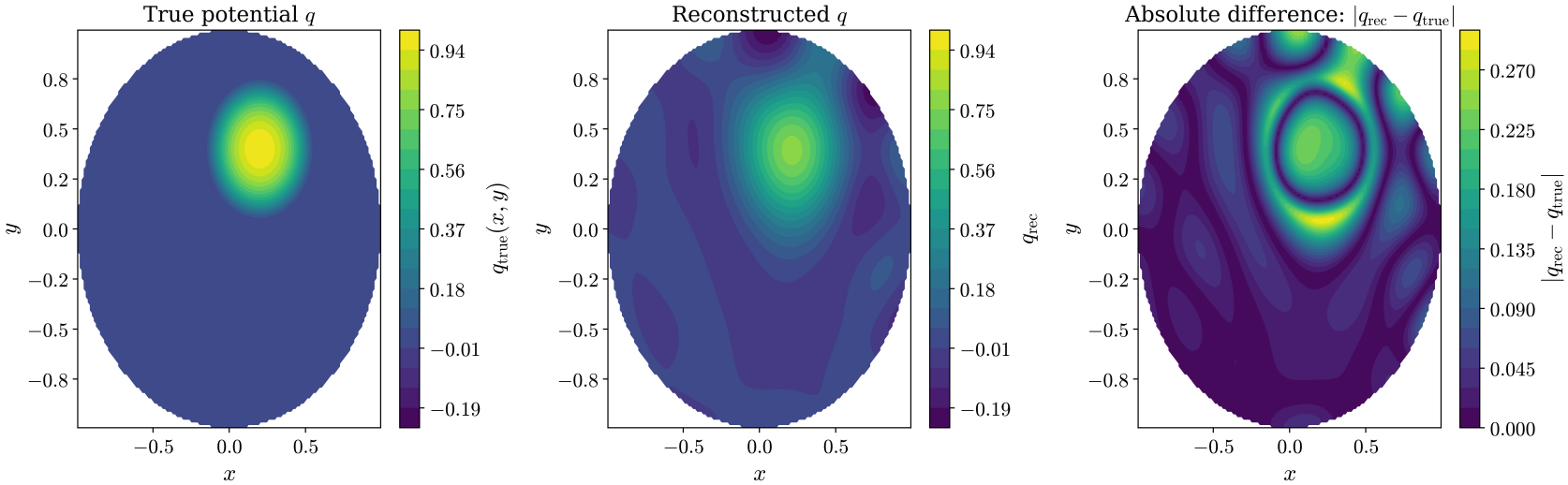


Centered bump reconstruction

Example 2: Off center bump

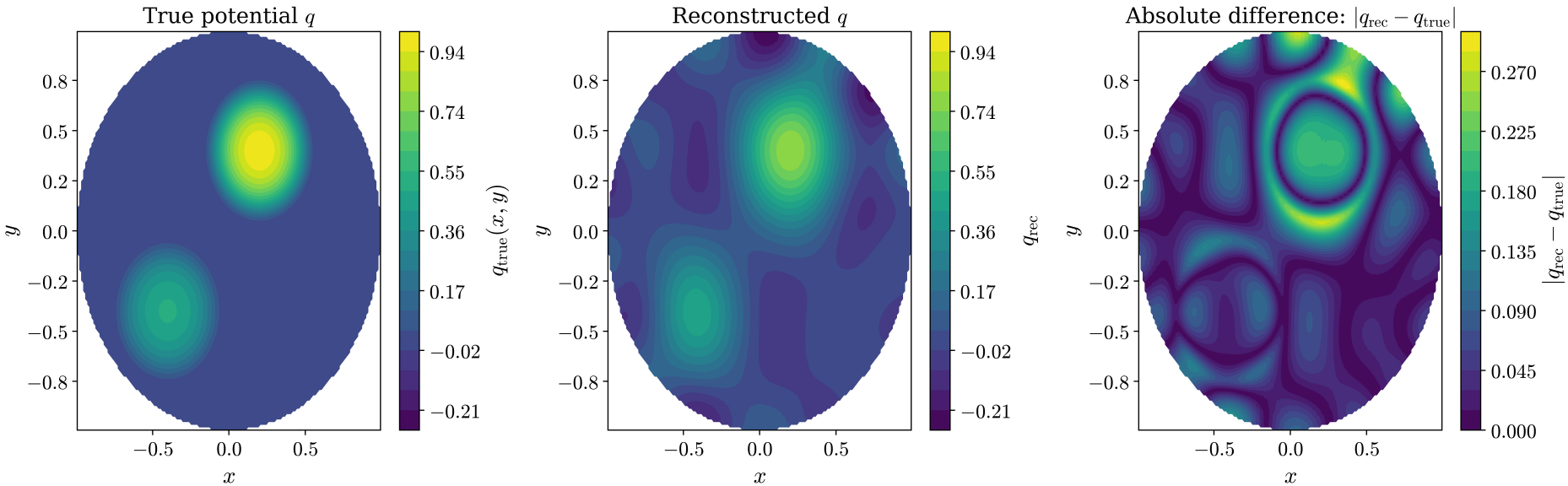


Recovered real Fourier data $\Re\mathcal{F}(q)$ for the off center bump.



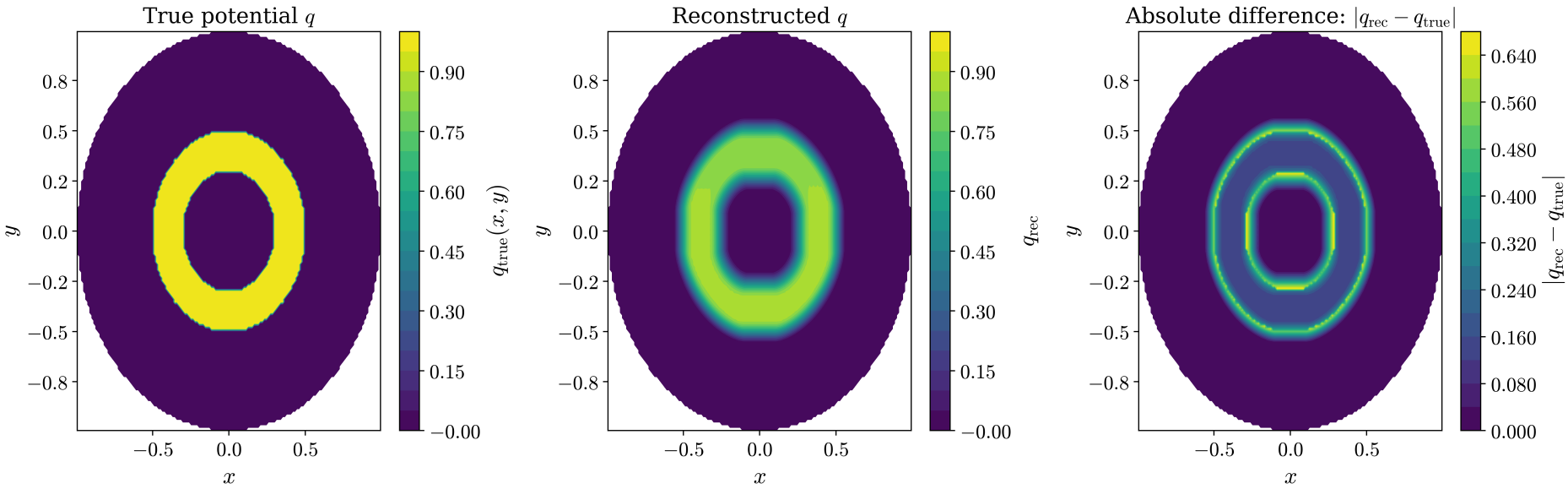
Off center bump reconstruction

Example 3: Two bumps



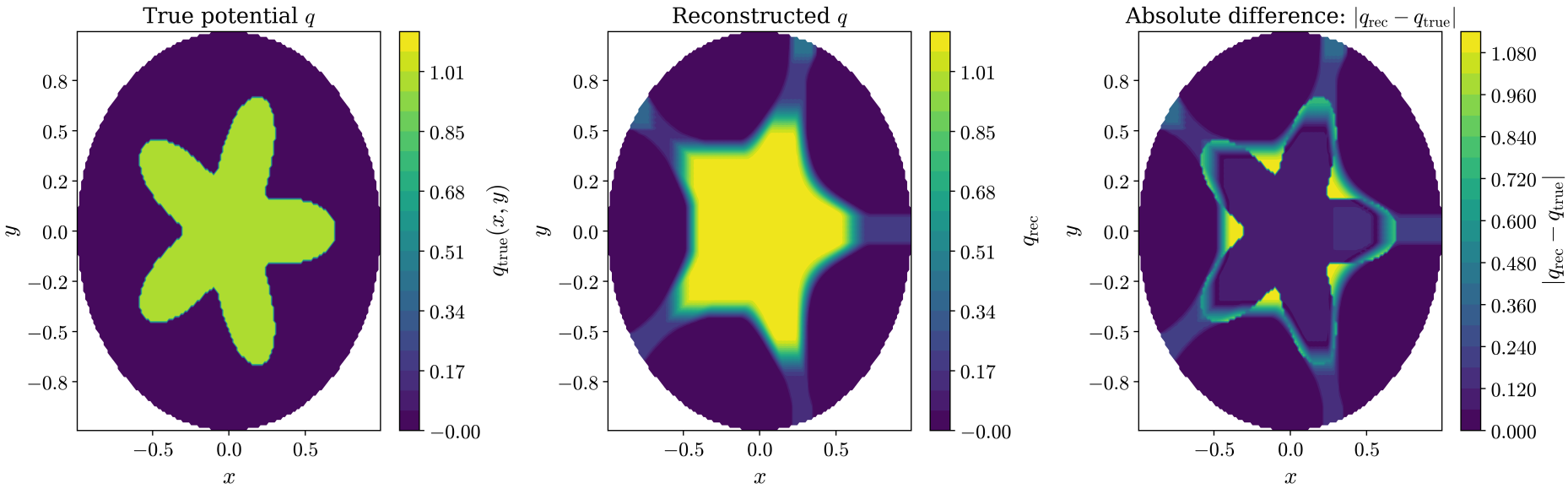
Two bump reconstruction

Example 4: Ring



Ring reconstruction

Example 5: Star



Ring reconstruction

Thank you for your attention!

UNIVERSITY OF OULU

Banach Fixed Point Theorem

Let (X, d) be a complete metric space, and let $T : X \to X$ be a **contraction mapping**, i.e., there exists a constant $0 \leq k < 1$ such that:

$$d(T(x), T(y)) \leq k \cdot d(x, y), \quad \forall x, y \in X.$$

Then:

1. The mapping T has a unique fixed point $x^* \in X$, i.e., $T(x^*) = x^*$.
2. For any initial point $x_0 \in X$, the sequence $\{x_n\}$ defined by:

$$x_{n+1} = T(x_n)$$

converges to x^*.